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Introduction to Statistical Methods

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Session 11

Correlation and Regression analysis

(Session 11: 17th / 18th January 2026)

Start Recording

Example 1

Suppose you define the time to get ready as the time (rounded to the nearest minute) from when you get out of bed to when you leave your home. You collect the times shown below for 10 consecutive work days:

Day	1	2	3	4	5	6	7	8	9	10
Time (min)	39	29	43	52	39	44	40	31	44	35

How do you summarize the above data?

Example 2

The owner of a showtime movie theatre wishes to predict weekly gross revenue based on advertising expenditures. The sample data are as follows:

Sl. No	1	2	3	4	5	6	7	8
Weekly gross revenue ('1000s)	96	90	95	92	95	94	94	94
Television advertising ('1000s)	5.0	2.0	4.0	2.5	3.0	3.5	2.5	3.0
Newspaper advertising ('1000s)	1.5	2.0	1.5	2.5	3.3	2.3	4.2	2.5

Is summarization of the above data enough? Or do we need to look beyond just summerization?

In example 1, as the data relates to a single variable, only summarization with descriptive statistics is enough. But in example 2, as there are three variables, besides summarization by descriptive statistics, it is also required to find out whether, there is relation between the variables.

In summary statistics, one important measure besides mean is variance (or standard deviation)

Covariance

Likewise, in the pursuit of finding the relation between two or more variables, the statistical technique used for studying the existence and extent of relationship between the variables is through Covariance $Cov\ (X, Y) = E\left(X-\overline{X}\right)\left(Y-\overline{Y}\right)$

Covariance

Covariance signifies the direction of the linear relationship between the variables. By direction we mean if the *variables* are directly proportional or inversely proportional to each other. (Increasing the value of one variable might have a positive or a negative effect on the value of the other variable).

Covariance

The values of covariance can be any number between the two opposite infinities. Also, it's important to mention that covariance only measures how two variables change together, not the dependency of one variable on another one.

Covariance

The upper and lower limits for the covariance depend on the variances of the variables involved. These variances, in turn, can vary with the scaling of the variables. Even a change in the units of measurement can change the covariance. Thus, covariance is only useful to find the direction of the relationship between two variables and not the magnitude.

Covariance

If X_1 and X_2 are two variables, the covariance between X_1 and X_2 is defined as

$$\mathrm{Cov}(X_1, X_2) = \frac{\sum_{i=1}^{n} (X_1 - \overline{X}_1)(X_2 - \overline{X}_2)}{n - 1}$$

Covariance

If $X_1, X_2, \ldots, X_n$ are n variables, then the Covariance between these n variables is the Variance – Covariance matrix given by

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdot & \cdot & \cdot & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \cdot & \cdot & \cdot & \sigma_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \sigma_{n1} & \sigma_{n2} & \cdot & \cdot & \cdot & \sigma_{nn} \end{bmatrix}$$

where

$$s_{ij} = \begin{cases} \text{Variance}, & \text{for } i = j \\ \text{Covariance}, & \text{for } i \neq j \end{cases}$$

All diagonal elements in the variance-covariance matrix are variances and off diagonal elements are covariances.

- is a symmetric matrix.

Covariance

Even though the covariance explains the relationship between two or more variables, the units the measurement of variable is attached with the value of the covariance. If they have different unit of measurements, based on the sign it is possible to say whether they are related or not, but the degree (strength) of relationship is not possible to decide based on the value of covariance due to different units of measurement.

Covariance

Example: Find the covariance between age (months) and weight (kgs) for the following data

9	6	5	8	6	7	Age (years)
13	11	10	12	8	12	Weight (kgs)

Covariance

Age (yrs) (X)	Weight (kgs) (Y)	$(X-\bar{X})$	$(Y-\bar{Y})$	$(X-\bar{X})(Y-\bar{Y})$
5	20	-2	-3	6
4	18	-3	-5	15
7	23	0	0	0
10	32	3	9	27
8	26	1	3	3
8	25	1	2	2
å $X = 42$	$\sum Y = 144$			å $(X-\bar{X})(Y-\bar{Y}) = 53$

$$\text{Cov}(X, Y) = \frac{53}{5} = 10.6 \text{ years.kgs ®}$$ what is the interpretation of this value?

Covariance

Covariance based on Probability distribution.

If X and Y are two random variables with probability mass function p(x)/ probability density function f(x), then the covariance between X and Y

$$\mathbf{Cov\ (X, Y) = E[(X-\mu_X)(Y-\mu_Y)] = E(XY) - E(X)E(Y)}$$

where $E(X) = \mu_X$ and $E(Y) = \mu_Y$ are the means of X and Y.

Covariance

Application of Covariance.

1. Multivariate linear regression:

If there a dependent variable and n independent variables, to find out the relationship between these (n+1) variables, a variance – covariance matrix will be helpful.

Covariance

Application of Covariance.

2. Time series:

In the given series of data over the time, it helps in finding the autocorrelation at different time lags.

For example, if $X_{t1}, X_{t2}, \ldots, X_{tn}$ denote the time series data over a given period, then the autocovariance

Covariance

Application of Covariance.

at lag h is given by

$$\gamma_X(t+h, t) = \mathrm{Cov}\,(X_{t+h}, X_t)$$

For example, if $X_{t1}, X_{t2}, \ldots, X_{tn}$ denote the time series data over a given period, then the auto-covariance.

The auto-covariance at lag h is then given by

$$\hat{\gamma}(h) = \frac{1}{n}\sum_{i=1}^{n-|h|}(X_{t+|h|} - \overline{X})(X_t - \overline{X}), -n < h < n.$$

Application of Covariance

With the help of auto-covariance, the autocorrelation is given by

$$r_k = \frac{\sum_{t=k+1}^{T} (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^{T} (y_t - \bar{y})^2}$$

Application of Covariance

2. Principal Component Analysis (PCA):

PCA is simply described as "diagonalizing the covariance matrix". What does diagonalizing a matrix mean in this context? It simply means that we need to find a non-trivial linear combination of our original variables such that the covariance matrix is diagonal.

Application of Covariance

When this is done, the resulting variables are uncorrelated, i.e. independent. The computation of the linear combinations of the original variables, called the ’principal components’.

Let A be a variance – covariance matrix and X be a vector of variables the Ax = ✦x, where ✦ is a scalar.

Application of Covariance

Calculate the eigenvalues and the corresponding eigenvectors of the covariance of the matrix A. If an eigenvalue computed in PCA is large, then the corresponding eigenvector (principal component) is important in describing the underlying data dependencies.

Application of Covariance

If the eigenvalue is near zero, that principal component is relatively unimportant and the data depend primarily on fewer components than there are measurement types. We merely need the machinery of eigenvectors and covariance matrices in order to find linear combinations of our original variables that are uncorrelated.

Correlation coefficient

To overcome the issue related to Covariance, Karl Pearson's suggestion was to divide the covariance of the variables by their respective standard deviations. The ratio of covariance to the product of standard deviations is called Karl Pearson's product moment correlation, which is free of unit of measurement.

Correlation coefficient

Karl Pearson's product moment correlation returns a numerical value, popularly known as **Coefficient of Correlation**, denoted by **'r'**

Correlation coefficient is a statistic that assesses the **strength** and **direction of relationship** of two continuous variables

Correlation coefficient

Types of Correlation

- Simple correlation
- Partial correlation
- Multiple correlation

Correlation coefficient

- Mathematically the following measures are used to find out the degree of relationship between the variables
- ♦ Karl Pearson's product moment correlation
- ♦ Spearman's rank correlation
- The numerical value obtained by the above methods is called correlation coefficient

Simple correlation coefficient

- Karl Pearson's product moment correlation or product moment correlation is also called simple correlation coefficient, denoted by 'r'

  It measures the nature and strength of relationship between two variables of the quantitative type

Karl Pearson's Correlation coefficient

- Formula for computing Karl Pearson's correlation coefficient (r) is standardised covariance (unit less)

$$r = \frac{\text{Covariance}(x, y)}{\sqrt{\text{Var}(x)}\sqrt{\text{Var}(Y)}}$$

Karl Pearson's Correlation coefficient

$$r = \frac{\dfrac{\sum_{i=1}^{n}(X-\overline{X})(Y-\overline{Y})}{n-1}}{\sqrt{\dfrac{\sum_{i=1}^{n}(X-\overline{X})^2}{n-1}}\sqrt{\dfrac{\sum_{i=1}^{n}(Y-\overline{Y})^2}{n-1}}} = \frac{\sum_{i=1}^{n}(X-\overline{X})(Y-\overline{Y})}{\sqrt{\sum_{i=1}^{n}(X-\overline{X})^2}\sqrt{\sum_{i=1}^{n}(Y-\overline{Y})^2}}$$

Karl Pearson's Correlation coefficient

In general for computation purpose the following formula can be used

$$r = \frac{n\sum XY - \left(\sum X\right)\left(\sum Y\right)}{\sqrt{n\sum X^2 - \left(\sum X\right)^2}\sqrt{n\sum Y^2 - \left(\sum Y\right)^2}}$$

Karl Pearson's Correlation coefficient

- The <u>sign</u> of 'r' denotes the nature of relationship
- The <u>value</u> of 'r' denotes the strength of relationship.

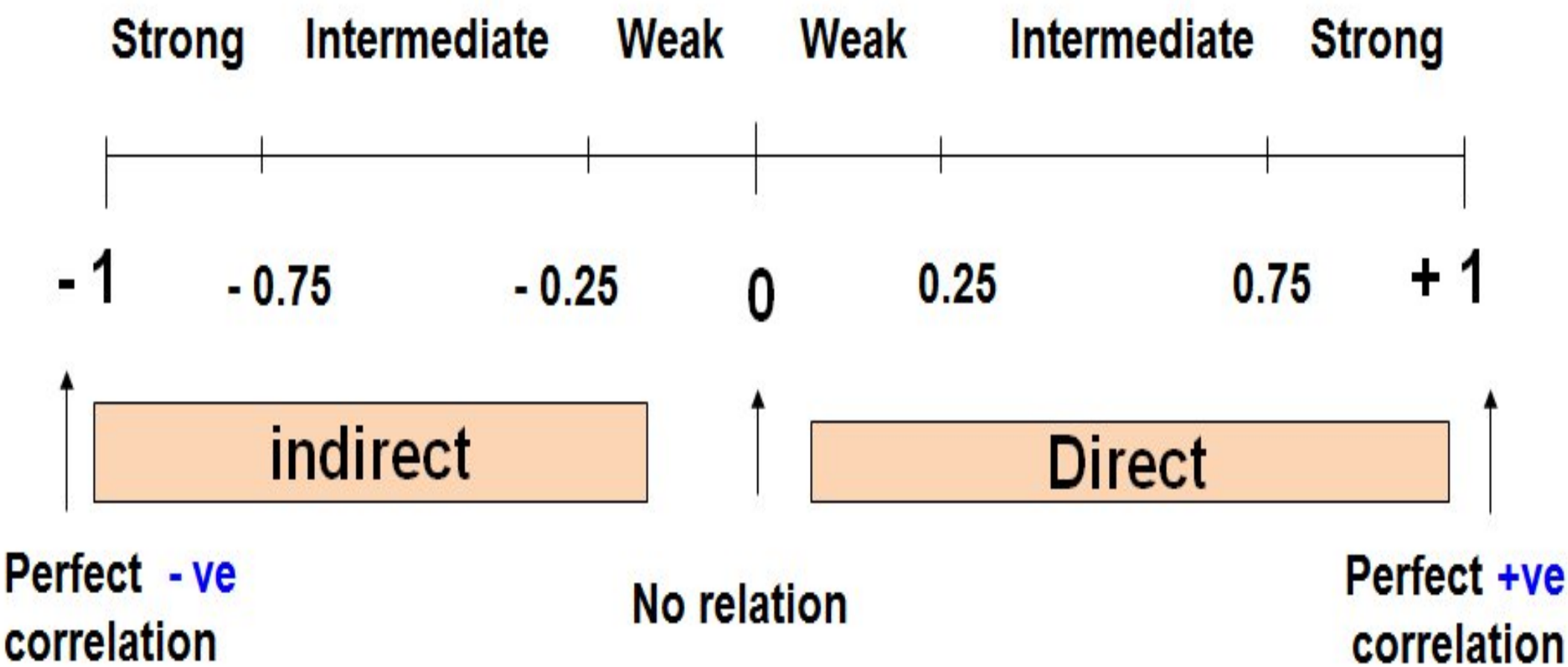
Karl Pearson's Correlation coefficient

- If the sign is +ve, this means the relation is direct (an increase in one variable is associated with an increase in the other variable and a decrease in one variable is related with a decrease in the other variable).
- If the sign is -ve this means an inverse or indirect relationship (an increase in one variable is related with a decrease in the other).

Karl Pearson's Correlation coefficient

- The value of r ranges between (-1) and (+1), i.e., $-1 \leq r \leq +1$
- The value of r denotes the strength of the relationship as illustrated by the following diagram.

Karl Pearson's Correlation coefficient



Karl Pearson's Correlation coefficient

- r = - 1 ⇒ Perfect negative correlation
- - 0.99 to - 0.76 ⇒ Strong negative correlation
- - 0.75 to - 0.26 ⇒ Intermediate (Moderate) negative correlation
- - 0.25 to 0 ⇒ Weak negative correlation
- r = 0 ⇒ Zero correlation
- 0 to 0.25 ⇒ Weak positive correlation
- 0.26 to 0.75 ⇒ Intermediate (Moderate) positive correlation
- 0.76 to 0.99 ⇒ Strong positive correlation
- r = + 1 ⇒ Perfect positive correlation

Karl Pearson's Correlation coefficient

Weight (Kg)	Age (yrs)	Serial No
12	7	1
8	6	2
12	8	3
10	5	4
11	6	5
13	9	6

Karl Pearson's Correlation coefficient

Y^2	X^2	XY	Wt (Kg) (Y)	Age (yrs) (X)	Sl no.
144	49	84	12	7	1
64	36	48	8	6	2
144	64	96	12	8	3
100	25	50	10	5	4
121	36	66	11	6	5
169	81	117	13	9	6
$\sum Y^2 = 742$	$\sum X^2 = 291$	$\sum XY = 461$	$\sum Y = 66$	$\sum X = 41$	Total

Karl Pearson's Correlation coefficient

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{n\sum X^2 - (\sum X)^2}\sqrt{n\sum Y^2 - (\sum Y)^2}}$$

$$r = \frac{6*461 - 41*66}{\sqrt{6*291 - (41)^2}\sqrt{6*742 - (66)^2}}$$

r = 0.759 (Positive (Direct) strong correlation)

Karl Pearson's Correlation coefficient

Example: Relationship between anxiety and test Scores

Anxiety (X)	Test score (Y)	X^2	Y^2	XY
10	2	100	4	20
8	3	64	9	24
2	9	4	81	18
1	7	1	49	7
5	6	25	36	30
6	5	36	25	30
$\sum X = 32$	$\sum Y = 32$	$\sum X^2 = 230$	$\sum Y^2 = 204$	$\sum XY = 129$

Karl Pearson's Correlation coefficient

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{n\sum X^2 - (\sum X)^2}\sqrt{n\sum Y^2 - (\sum Y)^2}}$$

$$r = \frac{6*129 - 32*32}{\sqrt{\left(6*230-(32)^2\right)\left(6*204-(32)^2\right)}} = \frac{774-1024}{\sqrt{(356)(200)}} = -0.94$$

r = - 0.94 (Negative (Indirect) strong correlation)

Spearman's Rank Correlation coefficient

- It is a non-parametric measure of correlation.
- This procedure makes use of the two sets of ranks that may be assigned to the sample values of x and Y.
- Spearman Rank correlation coefficient could be computed in the following cases:

Spearman's Rank Correlation coefficient

- Both variables are quantitative.
- Both variables are qualitative ordinal.
- One variable is quantitative and the other is qualitative ordinal.

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Spearman's Rank Correlation coefficient

- Rank the values of X with the smallest value as rank 1,
- next value as rank 2 and so on,
- assign rank n for the largest value.
- If there any tied values assign average ranks.
- The ranks can be assigned in the reverse order
- Rank the values of Y as done for X separately.

Spearman's Rank Correlation coefficient

- Compute the value of d_i for each pair of observation by subtracting the rank of Y_i from the rank of X_i
- Square each d_i and compute $\sum d^2_i$ which is the sum of the squared values.

$$R = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2 - 1)}$$

Where $d = R_x - R_y$, R_x = Rank of X values; R_y Rank of Y values

The value of R denotes the magnitude and nature of relationshion giving the same interpretation as simple r.

Spearman's Rank Correlation coefficient

Example

In this example, we will use the raw data in the table below to calculate the correlation between the IQ of a person with the number of hours spent in front of TV per week

Intelligent Quotient	106	86	100	101	99	103	97	113	112	110
Hours of TV watching	7	0	27	50	28	29	20	12	6	17

Spearman's Rank Correlation coefficient

Intelligent Quotient (X)	Hours of TV watching (Y)	Rank X (R_x)	Rank Y (R_y)	$d = R_x - R_y$	d^2
86	0	1	1	0	0
97	20	2	6	− 4	16
99	28	3	8	− 5	25
100	27	4	7	− 3	9
101	50	5	10	− 5	25
103	29	6	9	− 3	9
106	7	7	3	4	16
110	17	8	5	3	9
112	6	9	2	7	49
113	12	10	4	6	36

Spearman's Rank Correlation coefficient

With d_i^2 found, we can add them to find $\sum d_i^2 = 194$. The value of *n* is 10. So these values can now be substituted back into the equation, which evaluates to

$$R = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \text{ x } 194}{10(10^2 - 1)} = -0.176$$

This low value shows that there is a negative correlation between IQ and hours spent watching TV is very low.

Scattered diagram

- Scattered diagram is one of the ways of finding the extent of relationship between two quantitative variables. However, this method will only indicate that there is a relationship between two variables but, not the extent to which they are related.

Scattered diagram

- ♣ Rectangular coordinate
- ♣ Two quantitative variables
- ♣ One variable is called independent (X) and the
- second is called dependent (Y)
- ♣ Points are not joined
- ♣ No frequency table

Fig 1: Perfect negative correlation: r = – 1

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Fig 2: Strong negative correlation: $-0.99 \leq r < -0.75$

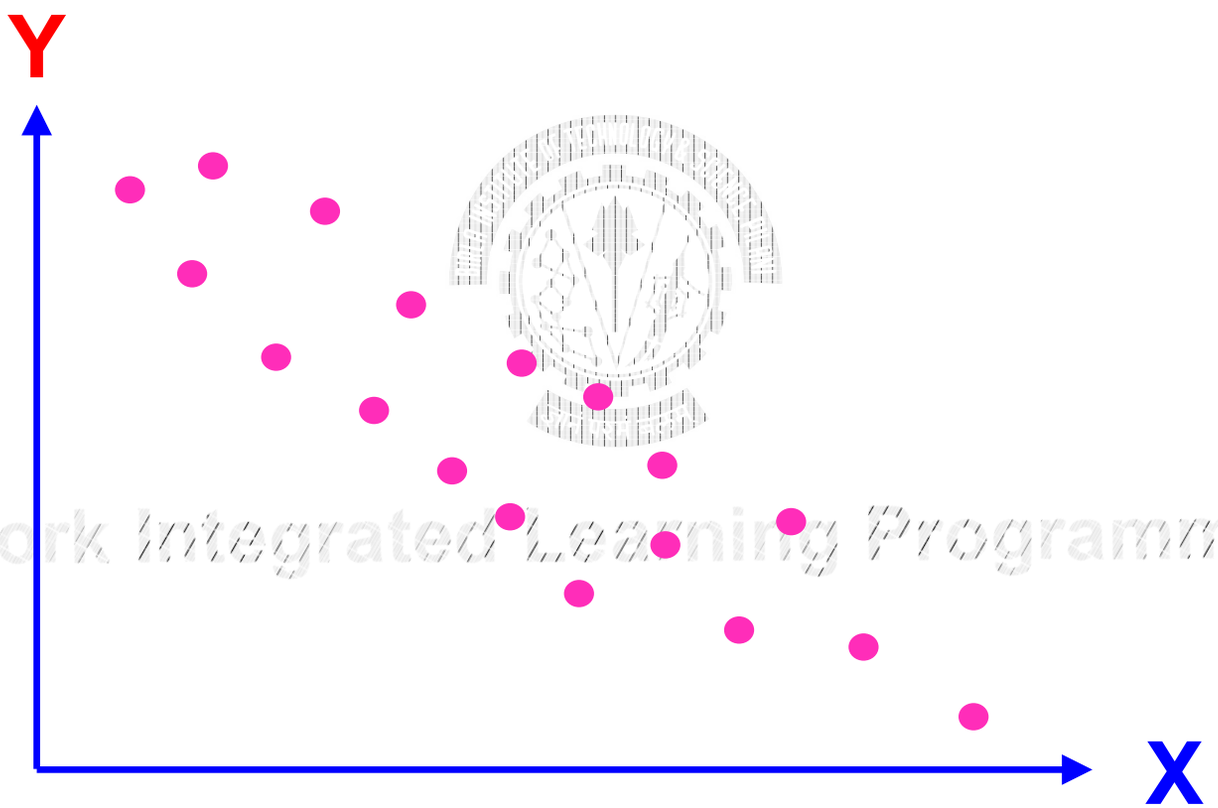

Fig 4: Moderate negative correlation: $-0.75 \leq r < -0.25$

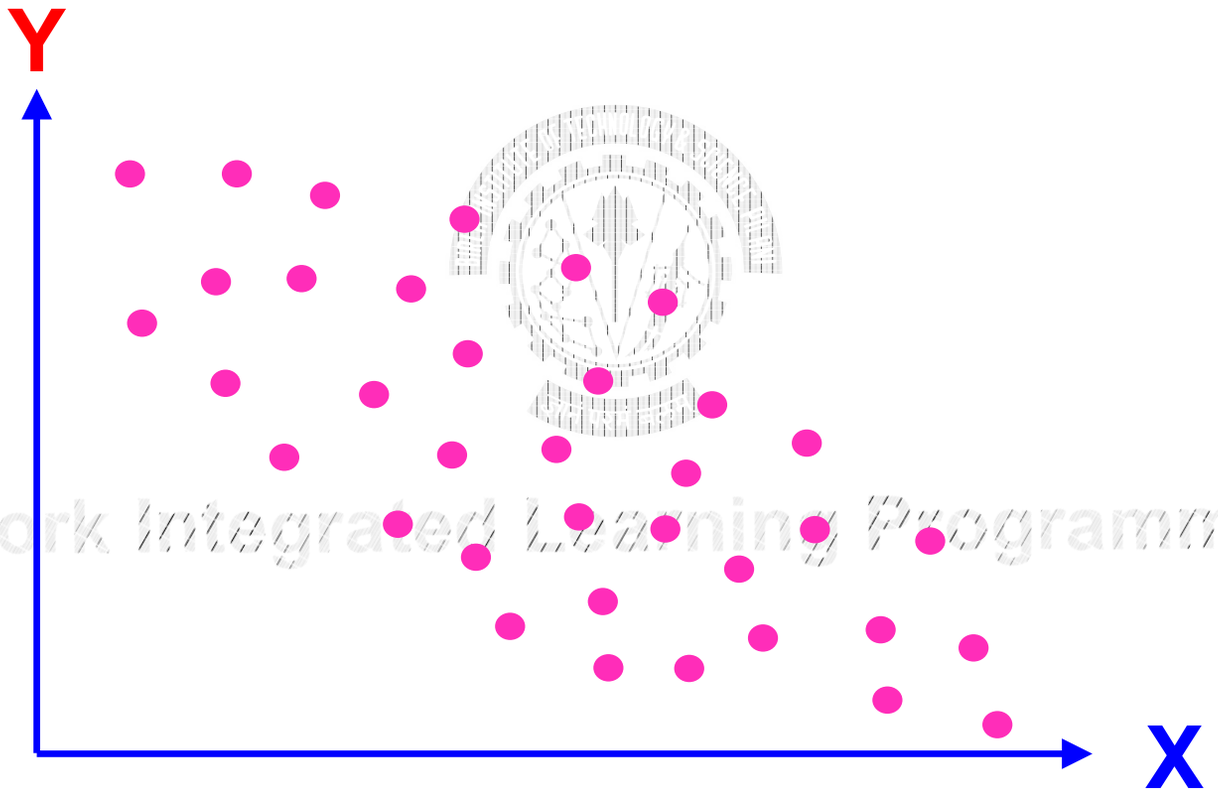

Fig 4: Weak negative correlation: $-0.25 \leq r < 0$

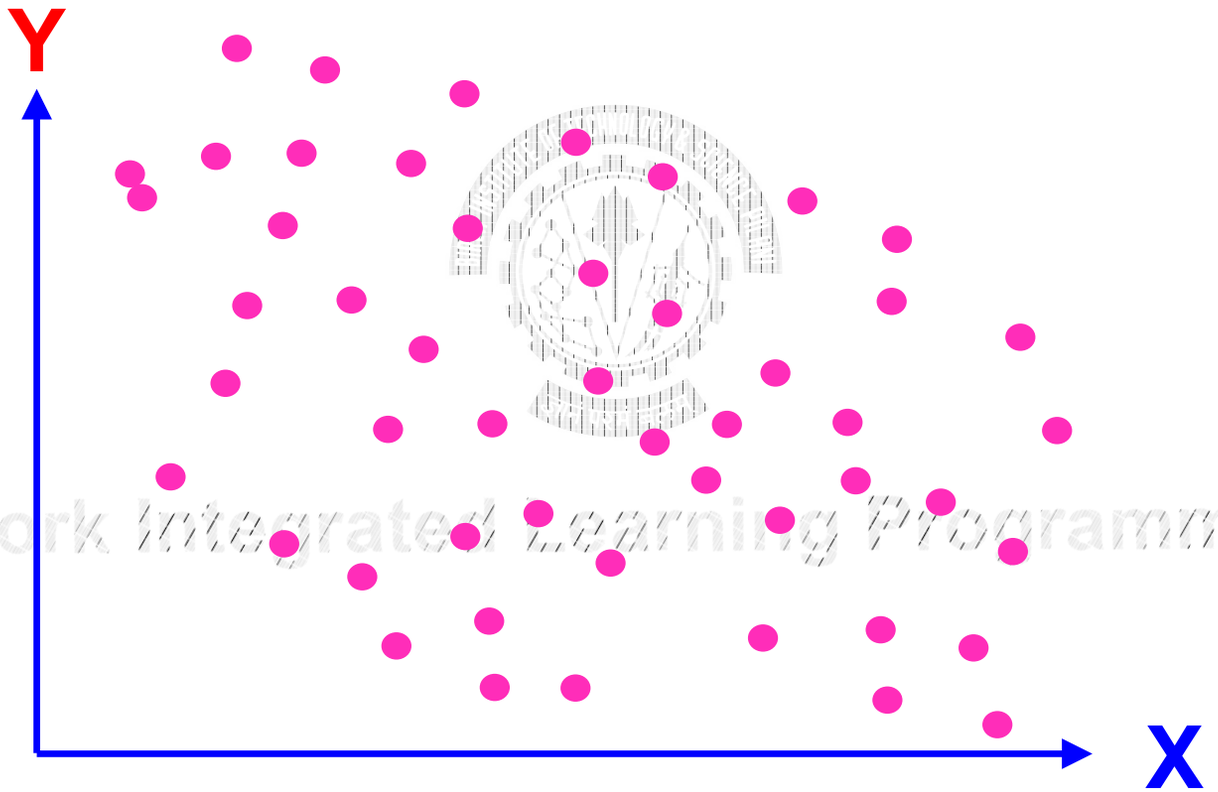

Fig 5 (a): Zero correlation: r = 0

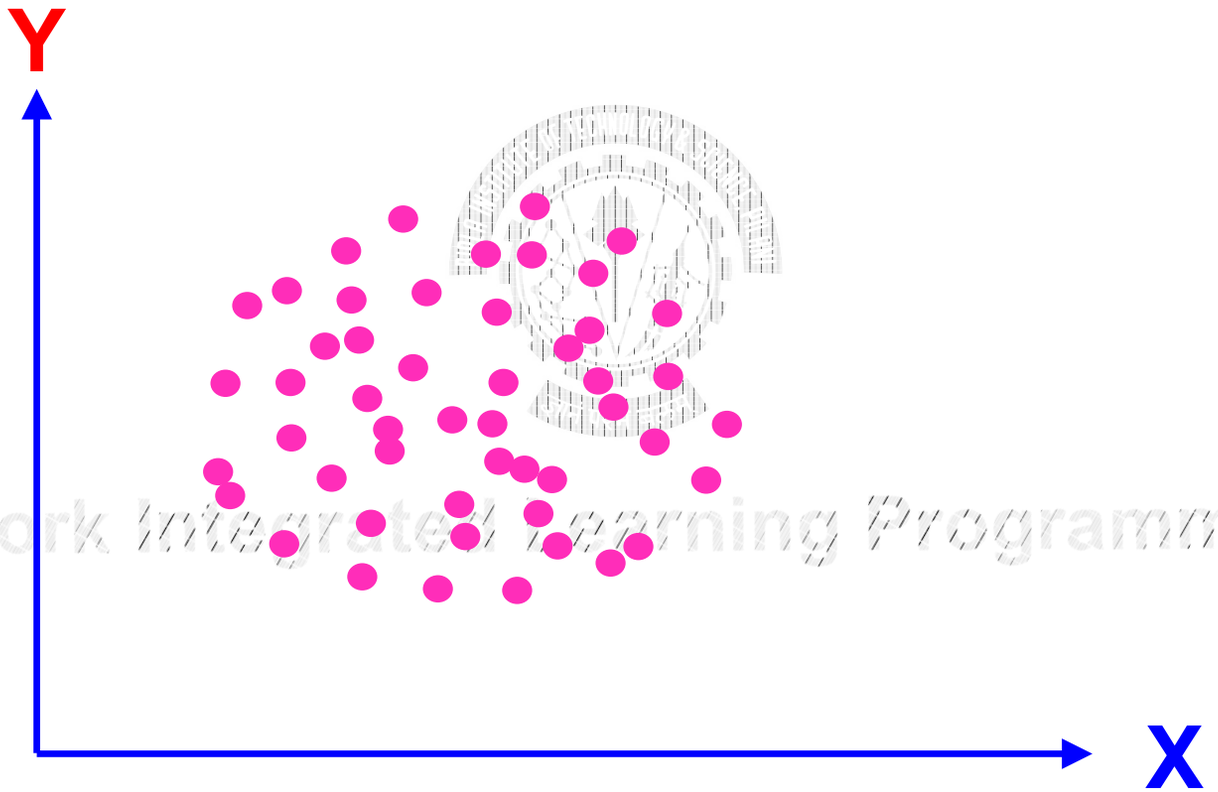

Fig 5 (b): Zero correlation: r = 0

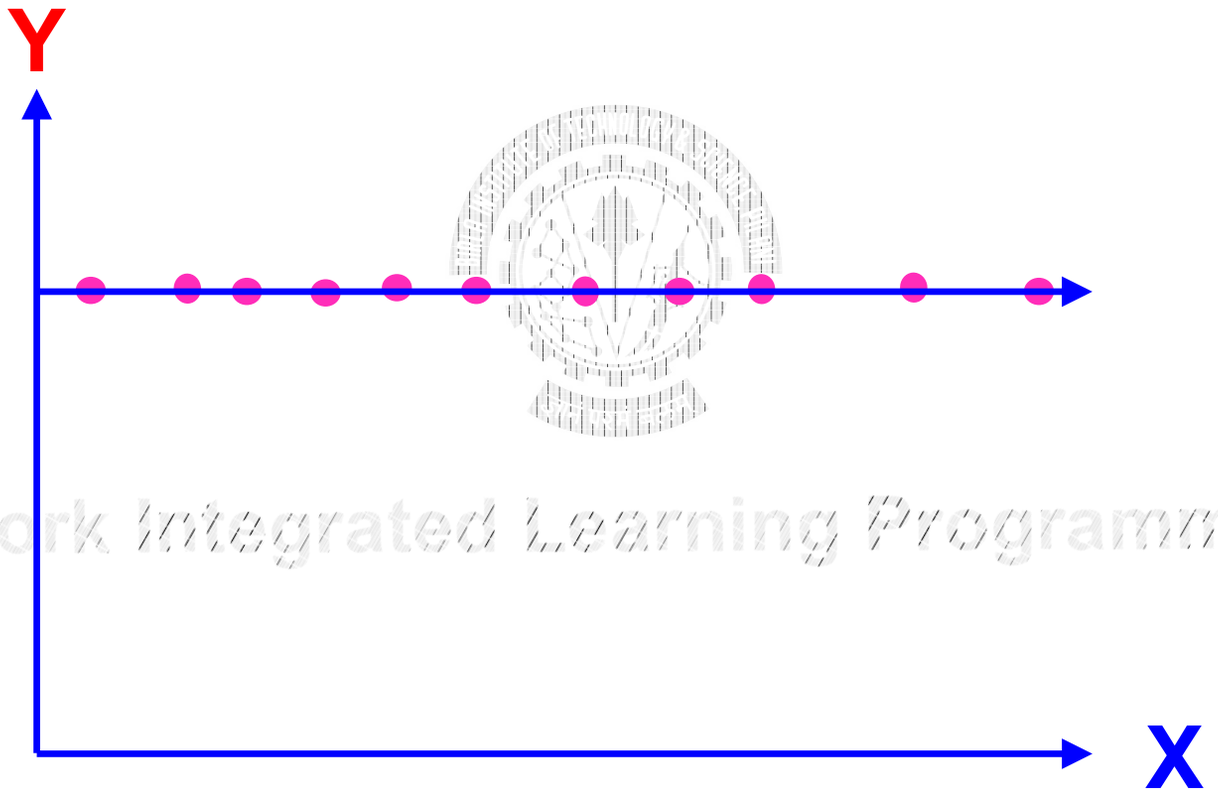

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Fig 5 (c): Zero correlation: r = 0

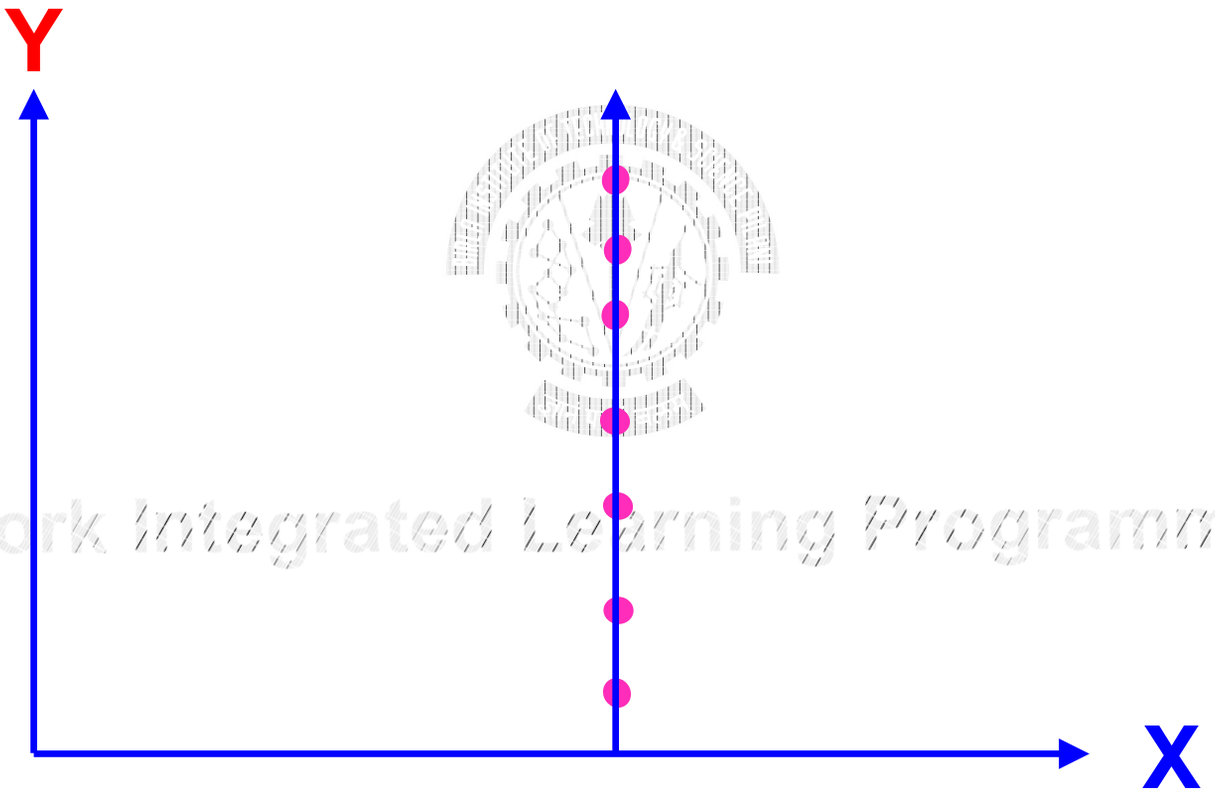

Fig 6: Weak positive correlation: $0 < r \leq 0.25$

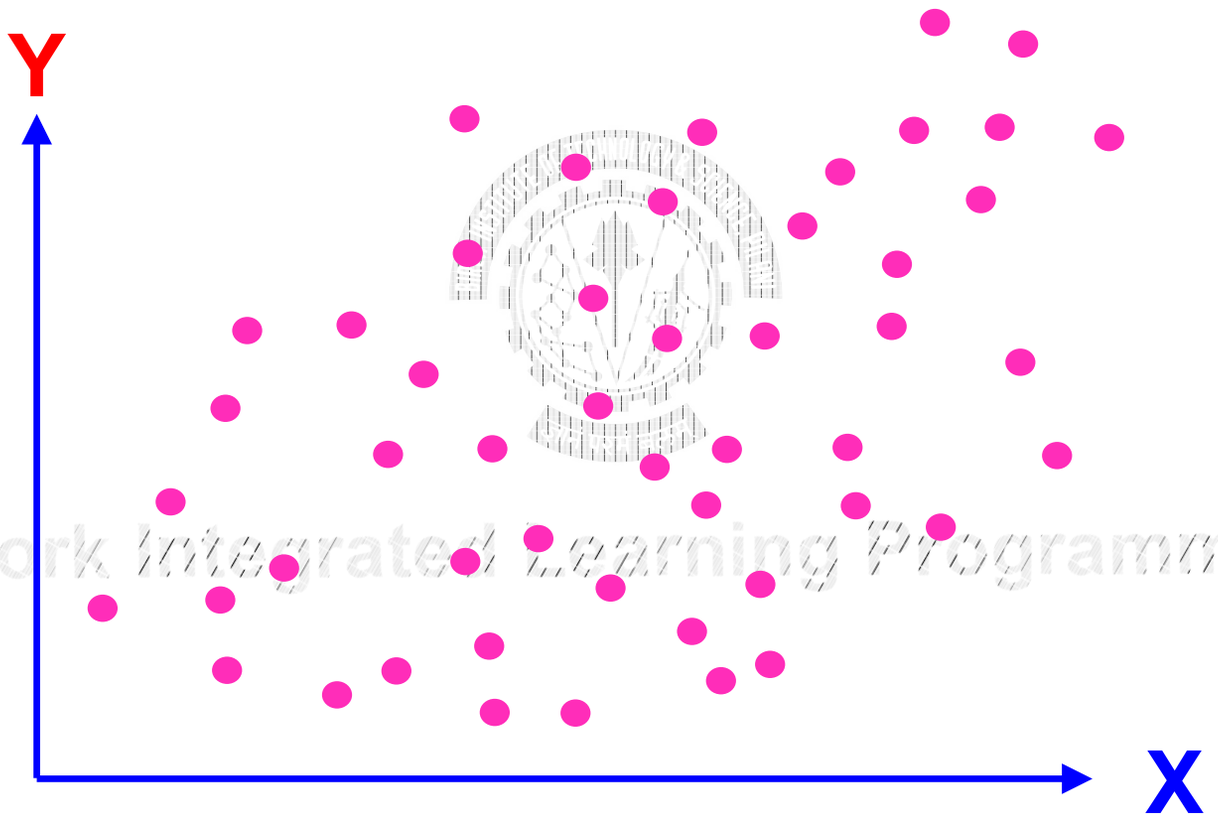

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Fig 7: Moderate positive correlation: $0.25 < r \leq 0.75$

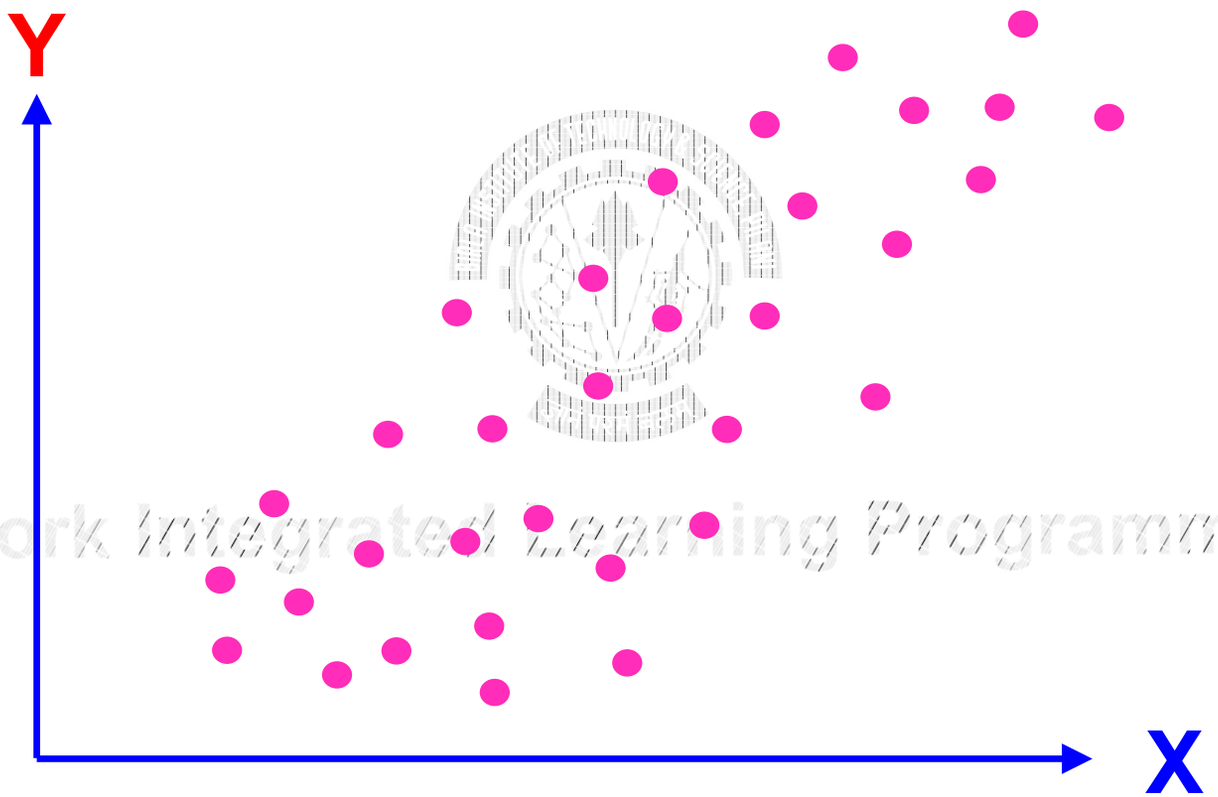

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Fig 8: Strong positive correlation: $0.75 < r \leq 0.99$

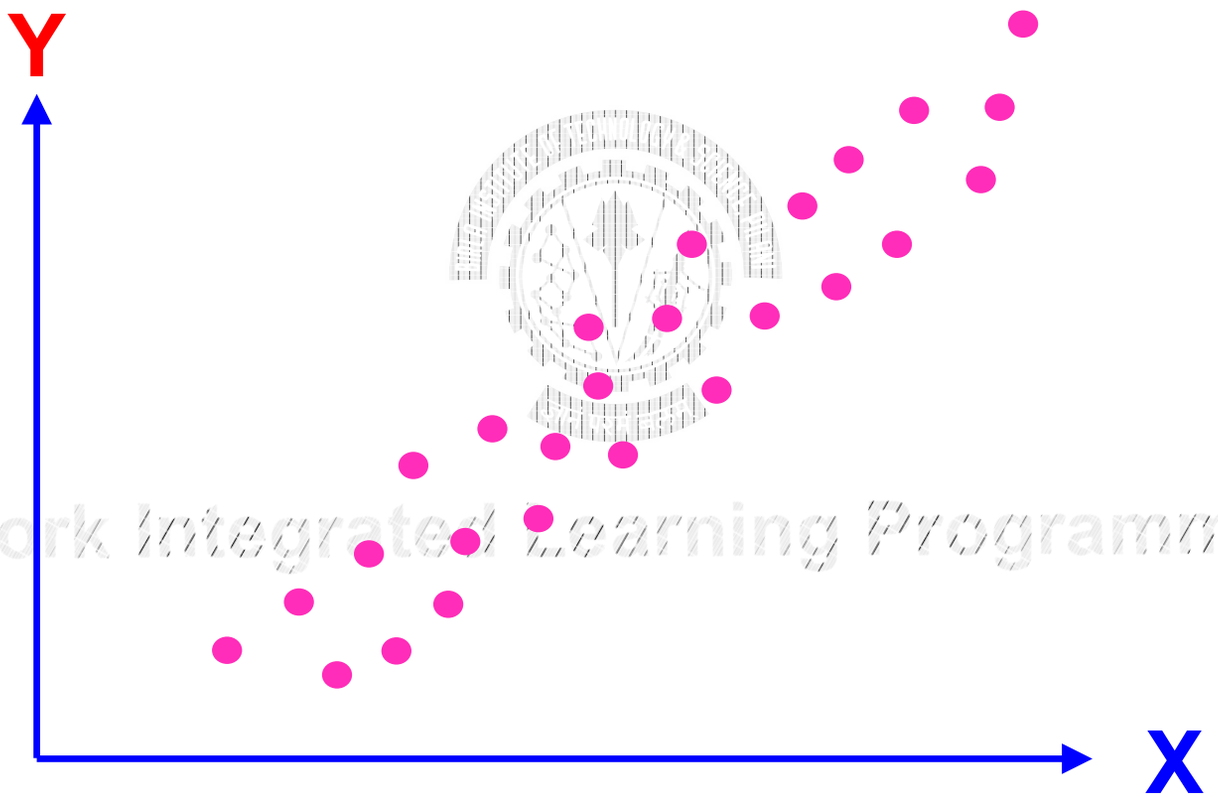

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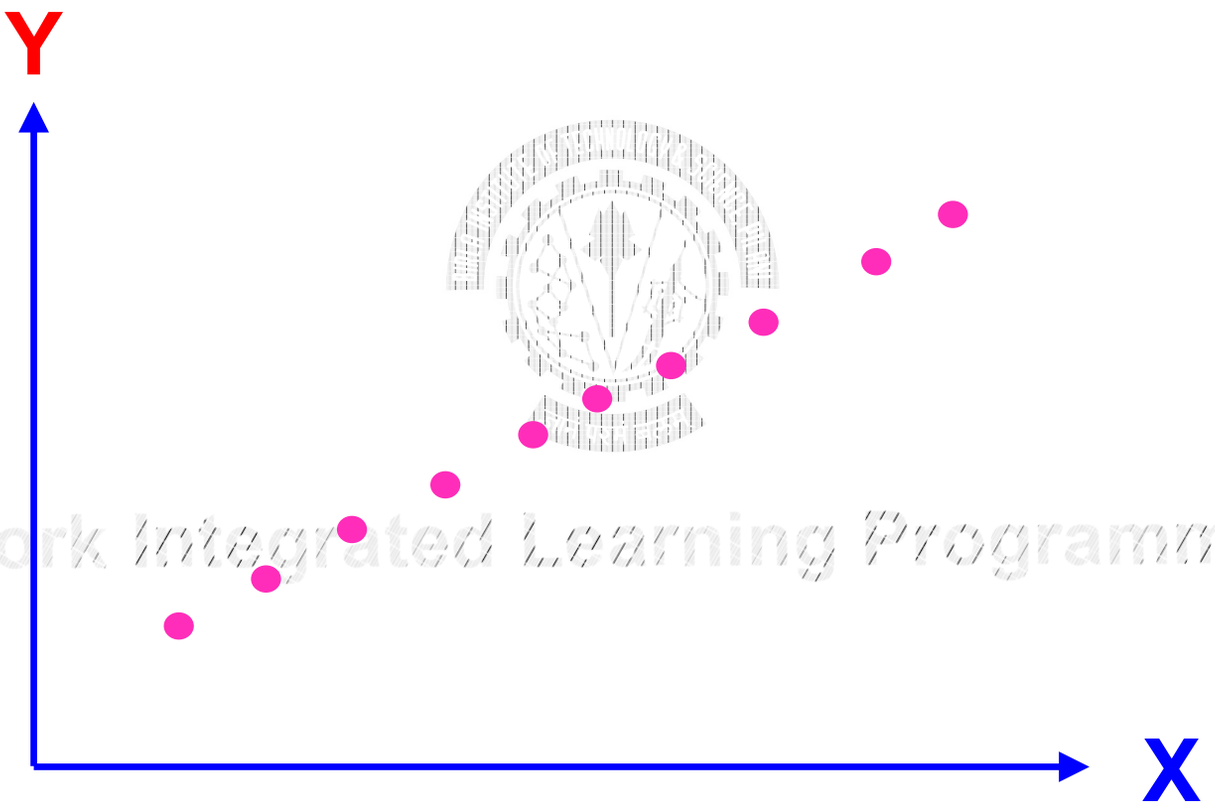
Fig 9: Perfect negative correlation: r = + 1



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Scattered diagram

The pattern of data is indicative of the type of relationship between your two variables:

- Positive relationship
- Negative relationship
- No relationship

Example 2

The owner of a showtime movie theatre wishes to predict weekly gross revenue based on advertising expenditures. The sample data are as follows:

Sl. No	1	2	3	4	5	6	7	8
Weekly gross revenue ('1000s)	96	90	95	92	95	94	94	94
Television advertising ('1000s)	5.0	2.0	4.0	2.5	3.0	3.5	2.5	3.0

Is summarization of the above data enough? Or do we need to look beyond just summerization?

Regression Analysis

Weekly gross revenue (Y) = Television advertising (X)

That is $$Y = X$$

But TV advertising will not be constant all the week. By adding a weighting factor to X to account for variability, the model becomes

$$Y = bX$$

Weekly gross revenue may not depend only on TV ad, but other factors may also act in increasing the revenue. Now, by adding 'a' as an anchor point, the modified model will be

Regression Analysis

$$Y = a + bX$$

While making measurements on Y and X, the possibility of error due to randomness has to be considered, thereby the full the model becomes

$$Y = a + bX + e$$

This model is called as simple linear regression model which can be used for predicting the weekly gross revenue based on Television advertisement.

Regression Analysis

Relationships:

•In science, we frequently measure two or more variables on the same individual (case, object, etc). We do this to explore the nature of the relationship among these variables. There are two basic types of relationships.

- Cause-and-effect relationships.
- Functional relationships

Simple Linear Regression

Relationships:

•**Function**: a mathematical relationship enabling us to predict what values of one variable (Y) correspond to given values of another variable (X).

• Y: is referred to as the *dependent variable*, the *response variable* or the *predicted variable*.

• X: is referred to as the *independent variable*, the *explanatory variable* or the *predictor variable*.

Simple Linear Regression

Do heavier people burn more energy?

This questions reflect a desire to understand the relationship between two variables

What we need:

1. A plot/ graph to view the relationship
2. Characteristic to describe
3. Measures of the characteristics
4. Method to make inferences about the relationship

Simple Linear Regression

Interpreting…characteristics to look for:

•Patterns:

- Form (clusters, scatter, linear..)
- Direction (positive, negative)
- Strength (how closely points follow form)

•Deviations:

- Outliers

Simple Linear Regression

- Regressions are still focuses on relationship, not causation.
- Relationship is a necessary prerequisite for inferring causation, but also:

1. The independent variable must preceded the dependent variable in time.
2. The two variables must be plausibly lined by a theory,
3. Competing independent variables must be eliminated.

Simple Linear Regression

Dependent Variable	Independent Variable: Nominal	Independent Variable: Interval - Ratio
Nominal	Considers the distribution of one variable across the categories of another variable	Considers how a change in a variable affects a discrete outcome
Interval - Ratio	Considers the difference between the mean of one group on a variable with another group	Considers the degree to which a change in one or two variables results in a change in another

Simple Linear Regression

Dependent Variable	Independent Variable: Nominal	Independent Variable: Interval - Ratio
Nominal	Chi-square test	Logistic Regression
Interval - Ratio	t-test, ANOVA, F-Test	Regression / Correlation

Simple Linear Regression

Population based model

Y-Intercept Slope Random Error

$$Y_i = \alpha + \beta X_i + e_i$$

Dependent (Response) Variable

Independent (Explanatory) Variable

Simple Linear Regression

Sample based model

Y-Intercept Slope Random Error

$$Y_i = a + b X_i + e_i$$

Dependent (Response) Variable

Independent (Explanatory) Variable

Simple Linear Regression

Model Assumptions about the error term Ɛ

1. The error Ɛ is a random variable with mean of zero.
2. The variance of Ɛ, denoted by σ^2, is the same for all values of the independent variable.
3. The values of Ɛ are independent.
4. The error Ɛ is a normally distributed random variable.

Simple Linear Regression

Least Squares (L-S): Minimize squared error.

$$\sum_{i=1}^{n}\left(Y_i-\hat{Y}_i\right)^2=\sum_{i=1}^{n}\hat{\varepsilon}_i^2$$

$$\sum_{i=1}^{n}\varepsilon_i^2=\sum_{i=1}^{n}(Y_i-a-bX_i)^2$$

$$\frac{\partial}{\partial a}\sum_{i=1}^{n}\varepsilon_i^2=\frac{\partial}{\partial a}\sum_{i=1}^{n}(Y_i-a-bX_i)^2=0$$

$$\sum_{i=1}^{n}Y_i=na-b\sum_{i-1}^{n}X_i\ldots\ldots\ldots\ldots\ldots(1)$$

Simple Linear Regression

Least Squares (L-S): Minimize squared error.

$$\sum_{i=1}^{n}\left(Y_i-\hat{Y}_i\right)^2=\sum_{i=1}^{n}\hat{\varepsilon}_i^2$$

$$\sum_{i=1}^{n}\varepsilon_i^2=\sum_{i=1}^{n}(Y_i-a-bX_i)^2$$

$$\frac{\partial}{\partial b}=\sum_{i=1}^{n}\varepsilon_i^{\,2}=\frac{\partial}{\partial b}\sum_{i=1}^{n}(Y_i-a-bX_i)^2=0$$

$$\sum_{i=1}^{n}Y_i=\left(a\sum_{i=1}^{n}X_i-b\sum_{i=1}^{n}X_i\right)(-2X_i)=0$$

Simple Linear Regression

$$\frac{\partial}{\partial b}\sum_{i=1}^{n}\varepsilon_i^2 = \frac{\partial}{\partial b}\sum_{i=1}^{n}(Y_i - a - bX_i)^2 = 0$$

$$\sum_{i=1}^{n} Y_i = \left(a\sum_{i=1}^{n} X_i - b\sum_{i=1}^{n} X_i\right)(-2X_i) = 0$$

$$\sum_{i=1}^{n} X_i Y_i = a\sum_{i=1}^{n} X_i + b\sum_{i=1}^{n} X_i^2 \ldots\ldots\ldots\ldots\ldots(2)$$

Equations (1) and (2) are called normal equations. Solving these as simultaneous equations, we get

Simple Linear Regression

$$\sum_{i=1}^{n} Y_i = na + b\sum_{i=1}^{n} X_i \ldots\ldots\ldots\ldots\ldots\ldots(1)$$

$$\sum_{i=1}^{n} X_i Y_i = a\sum_{i=1}^{n} X_i + b\sum_{i=1}^{n} X_i^{\,2} \ldots\ldots\ldots\ldots\ldots(2)$$

$$\hat{b} = \frac{n\sum_{i=1}^{n} X_i Y_i - \left(\sum_{i=1}^{n} X_i\right)\left(\sum_{i=1}^{n} Y_i\right)}{n\sum_{i=1}^{n} X_i^{\,2} - \left(\sum_{i=1}^{n} Xi\right)^2} = \boxed{\frac{\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})}{\sum_{i=1}^{n}(Xi - X)^2}}$$

$$\hat{a} = \overline{Y} - \hat{b}\overline{X}$$

The fitted regression model is

$$\hat{Y} = \hat{a} + \hat{b}X$$

The owner of a showtime movie theatre wishes to predict weekly gross revenue based on advertising expenditures. The sample data are as follows:

Sl. No	1	2	3	4	5	6	7	8
Weekly gross revenue ('1000s)	96	90	95	92	95	94	94	94
Television advertising ('1000s)	5.0	2.0	4.0	2.5	3.0	3.5	2.5	3.0

	Weekly gross revenue ('1000s) (Y)	Television advertising ('1000s) (X)	$x = (X - \overline{X})$	$y = (Y - \overline{Y})$	xy	x^2
	96	5.0	2.25	1.81	4.08	5.06
	90	2.0	-3.75	-1.19	4.45	14.06
	95	4.0	1.25	0.81	1.02	1.56
	92	2.5	-1.75	-0.69	1.20	3.06
	95	3.0	1.25	-0.19	-0.23	1.56
	94	3.5	0.25	0.31	0.08	0.06
	94	2.5	0.25	-0.69	-0.17	0.06
	94	3.0	0.25	-0.19	-0.05	0.06
Sum	750	25.5	0	0	**10.375**	**25.5**
Mean	**93.75**	**3.19**				

Simple Linear Regression

$$\hat{b} = \frac{\sum xy}{\sum x^2} = \frac{10.375}{25.5} = 0.407$$

$$\hat{a} = \overline{Y} - \hat{b} * \overline{X} = 93.75 - 0.407 * 3.15 = 92.45$$

$$\hat{Y} = \hat{a} + \hat{b} * X = 92.45 + 0.407X$$

When X = 4.5, $\hat{Y} = 92.45 + 0.407 * 4.5 = 94.28$

Interpretation of ‘b’

For every unit change in X, there will be ‘b’ times changes in Y.

Work Integrated Learning Programmes

Simple Linear Regression

Coefficient of Determination

- Relationship Among SST, SSR, SSE

$$SST = SSR + SSE$$

$$\sum (Y_i - \bar{Y})^2 = \sum (\hat{Y}_i - \bar{Y})^2 + \sum (Y_i - \hat{Y}_i)^2$$

where:

SST = Sum of Squares due to Total
SSR = Sum of Squares due to Regression
SSE = Sum of Squares due to Error

Simple Linear Regression

Coefficient of Determination

$$r^2 = \frac{SSR}{SST} = \frac{\sum(\hat{Y} - \bar{Y})^2}{\sum(Y - \bar{Y})^2}$$

where:

SST = Sum of Squares due to Total

SSR = Sum of Squares due to Regression

Simple Linear Regression

Example

No. of TV advs (X)	No. of cars sold (Y)
1	14
3	24
2	18
1	17
3	27
2	20

Fit a regression line of Y on X and determine the coefficient of determination

Simple Linear Regression

	X	Y	x = (X-2)	y = (Y-20)	xy	x^2	$(Y-20)^2$	Y^ = 10 + 5X	$(Y^\wedge - 20)^2$
	1	14	-1	-6	6	1	36	15	25
	3	24	1	4	4	1	16	25	25
	2	18	0	-2	0	0	4	20	0
	1	17	-1	-3	3	1	9	15	25
	3	27	1	7	7	1	49	25	25
Total	10	100	0	0	20	4	114		100

Regression y on X

$$\hat{Y} = 10 + 5X$$

Coefficient of Determination

$$r^2 = \frac{SSR}{SST} = \frac{\sum (\hat{Y} - \bar{Y})^2}{\sum (Y - \bar{Y})^2} = \frac{100}{114} 0.8772$$

Simple Linear Regression

SL. No	X	Y	x = (X-2)	y = (Y-20)	xy	x^2	$(Y-20)^2$	Y^ = 10 + 5X	$(Y^\wedge - 20)^2$
1	1	14	-1	-6	6	1	36	15	25
2	3	24	1	4	4	1	16	25	25
3	2	18	0	-2	0	0	4	20	0
4	1	17	-1	-3	3	1	9	15	25
5	3	27	1	7	7	1	49	25	25
Total	10	100	0	0	20	4	114		100

Regression y on X

$$\hat{Y} = 10 + 5X$$

Coefficient of Determination

$$r^2 = \frac{SSR}{SST} = \frac{\sum (\hat{Y} - \bar{Y})^2}{\sum (Y - \bar{Y})^2} = \frac{100}{114} 0.8772$$

Simple Linear Regression

Interpretation

The regression relationship is very strong; 87.72% of the variability in the number of cars sold can be explained by the linear relationship between the number of TV ads and the number of cars sold